Find the Error

Grade 6 · 6.RP.1

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

There are 3 red beads and 5 blue beads. What is the ratio of red to blue? Student's answer: 5:3

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **Red comes first in the question, so the ratio is red to blue: 3:5. The student reversed the order.**

Correct answer: 3:5

Find the Error

Grade 6 · 6.RP.1

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A class has 4 cats and 6 dogs as pets. Write the ratio of cats to dogs in simplest form. Student's answer: 4:6

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **Both numbers can be divided by 2. The simplest form is 2:3.**

Correct answer: **2:3**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A recipe uses 2 cups of flour for every 3 cups of sugar. Student says: For every 3 cups of flour, there are 2 cups of sugar.

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The ratio is flour to sugar, 2:3. For every 2 cups of flour there are 3 cups of sugar, not the other way.**

Correct answer: **For every 2 cups of flour, there are 3 cups of sugar**

Find the Error

Grade 6 · 6.RP.2

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A car drives 180 miles in 3 hours. What is its unit rate in miles per hour? Student's answer: 3

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student divided the time by the distance. Divide the distance by the time: 180 ÷ 3 = 60 miles per hour.**

Correct answer: **60 miles per hour**

Find the Error

Grade 6 · 6.RP.2

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Five pounds of apples cost $15. What is the price per pound? Student's answer: $10

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student subtracted 5 from 15. Divide: $15 ÷ 5 = $3 per pound.**

Correct answer: **$3 per pound**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Which is the better buy: 8 cans for $6 or 5 cans for $4? Student says: 5 cans for $4

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **8 cans for $6 is $0.75 per can. 5 cans for $4 is $0.80 per can. The lower unit price is the better buy.**

Correct answer: **8 cans for $6**

Find the Error

Grade 6 · 6.RP.3

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

3 bags cost $9 and 5 bags cost $15. What is the cost of 10 bags? Student's answer: $25

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **Each bag costs $3. Ten bags cost 10 × $3 = $30. The student added 10 to 15.**

Correct answer: **$30**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

What is 25% of 60? Student's answer: 6

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student found 10% instead of 25%. 25% of 60 is 0.25 × 60 = 15.**

Correct answer: 15

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

On a map, 1 inch equals 40 miles. How many miles are 2.5 inches? Student's answer: 80 miles

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student used 2 inches instead of 2.5. 2.5 × 40 = 100 miles.**

Correct answer: **100 miles**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Divide $\frac{2}{3} \div \frac{1}{4}$. Student's answer: $\frac{1}{6}$

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student multiplied the fractions. Multiply by the reciprocal of $\frac{1}{4}$, which is 4: $\frac{2}{3} \times 4 = \frac{8}{3} = 2\frac{2}{3}$.**

Correct answer: $\frac{8}{3}$ or $2\frac{2}{3}$

Find the Error

Grade 6 · 6.NS.1

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Divide $\frac{3}{4} \div 3$. Student's answer: $\frac{9}{4}$

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student multiplied by 3 instead of dividing. 3 is $\frac{3}{1}$, so its reciprocal is $\frac{1}{3}$. $\frac{3}{4} \times \frac{1}{3} = \frac{1}{4}$.**

Correct answer: $\frac{1}{4}$

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Divide $\frac{5}{6} \div \frac{2}{3}$. Student's answer: $\frac{5}{9}$

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student divided the numerators and the denominators. Multiply by $\frac{3}{2}$: $\frac{5}{6} \times \frac{3}{2} = \frac{15}{12} = \frac{5}{4} = 1\frac{1}{4}$.**

Correct answer: $\frac{5}{4}$ or $1\frac{1}{4}$

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Divide 4.8 ÷ 0.6. Student's answer: 0.8

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **Move the decimal one place in each number to get 48 ÷ 6 = 8.**

Correct answer: 8

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Divide 12.6 ÷ 0.7. Student's answer: 1.8

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student moved only one decimal point. Use 126 ÷ 7 = 18.**

Correct answer: 18

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Divide 36.75 ÷ 2.5. Student's answer: 1.47

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student moved the decimal point the wrong way. Use 367.5 ÷ 25 = 14.7.**

Correct answer: 14.7

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Add 3.25 + 4.8. Student's answer: 7.13

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student did not line up the decimals. Write 4.80 and add: 3.25 + 4.80 = 8.05.**

Correct answer: 8.05

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Subtract 9.4 - 2.67. Student's answer: 7.73

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **Write 9.40 and subtract. 9.40 - 2.67 = 6.73. The student made a subtraction error in the hundredths place.**

Correct answer: 6.73

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Multiply 2.5 × 4.6. Student's answer: 115

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **Two decimal places total, but the student did not place the decimal. 25 × 46 = 1,150, so 2.5 × 4.6 = 11.5.**

Correct answer: **11.5**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Find the greatest common factor of 24 and 36. Student's answer: 6

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **6 is a common factor, but it is not the greatest. The factors of 24 and 36 include 12, which is the greatest.**

Correct answer: **12**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Find the least common multiple of 6 and 8. Student's answer: 48

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **48 is a common multiple, but 24 is smaller. The least common multiple is 24.**

Correct answer: 24

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Use the distributive property: $4(3 + 5)$. Student's answer: 17

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student multiplied 4 × 3 and then added 5. Multiply 4 by both terms: 4 × 3 + 4 × 5 = 12 + 20 = 32.**

Correct answer: 32

Find the Error

Grade 6 · 6.NS.5

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

The temperature is -5°C. It rises 8 degrees. What is the new temperature? Student's answer: -13°C

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **Rising means moving up on the number line. -5 + 8 = 3°C.**

Correct answer: **3°C**

Find the Error

Grade 6 · 6.NS.5

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A bank account has $50. A withdrawal of $80 is made. What is the new balance? Student's answer: $30

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The account is overdrawn, so the balance is negative. $50 - $80 = -$30.**

Correct answer: **-$30**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Which is less: -3 or -7? Student's answer: -3

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **-7 is farther to the left on the number line, so it is less than -3.**

Correct answer: **-7**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

What is the opposite of $\frac{7}{2}$? Student's answer: $\frac{2}{7}$

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student flipped the fraction. The opposite of $\frac{7}{2}$ is $-\frac{7}{2}$.**

Correct answer: $-\frac{7}{2}$

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

In which quadrant is the point (-3, 4)? Student's answer: first quadrant

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The x-coordinate is negative and the y-coordinate is positive. That is the second quadrant.**

Correct answer: **Second quadrant**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

The point (2, -5) is reflected across the y-axis. What is the new point? Student's answer: (2, -5)

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **Reflecting across the y-axis changes the sign of x. The new point is (-2, -5).**

Correct answer: **(-2, -5)**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

What is the absolute value of -9? Student's answer: -9

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **Absolute value is distance from zero, which is never negative. |-9| = 9.**

Correct answer: 9

Find the Error

Grade 6 · 6.NS.7

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Order these numbers from least to greatest: -2, 1, -5, 0. Student's answer: -2, -5, 0, 1

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student ordered the negatives wrong. -5 is less than -2, so the order is -5, -2, 0, 1.**

Correct answer: **-5, -2, 0, 1**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Is -4 greater than |-6|? Student says: Yes

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **|-6| = 6, and 6 is greater than any negative number. So -4 is not greater.**

Correct answer: **No**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Find the distance from (0, 0) to (0, 7). Student's answer: 0

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The points are 7 units apart along the y-axis. The distance is 7.**

Correct answer: 7

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Find the distance between (2, 3) and (8, 3). Student's answer: 5

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student miscounted. 8 − 2 = 6, so the distance is 6 units.**

Correct answer: 6

Find the Error

Grade 6 · 6.NS.8

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Find the distance between (-4, 1) and (3, 1). Student's answer: 4

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student counted only from -4 to 0. Use 3 - (-4) = 7 units.**

Correct answer: 7

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Evaluate 4^3. Student's answer: 12

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student multiplied 4×3. $4^3 = 4 \times 4 \times 4 = 64$.**

Correct answer: **64**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Evaluate 2^5. Student's answer: 10

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student multiplied 2×5. $2^5 = 2 \times 2 \times 2 \times 2 \times 2 = 32$.**

Correct answer: 32

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Evaluate $3 + 2^2$. Student's answer: 25

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student squared 5 instead of 2. Only 2 is raised to the power: $2^2 = 4$, so $3 + 4 = 7$.**

Correct answer: 7

Find the Error

Grade 6 · 6.EE.2

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Write an expression for: 5 less than 3 times n. Student writes: 5 - 3n

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **3 times n is 3n, and 5 less than that is 3n - 5. The order of the words matters.**

Correct answer: **3n - 5**

Find the Error

Grade 6 · 6.EE.2

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Evaluate 2a + 3b when a = 4 and b = 5. Student's answer: 14

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student added all the numbers. Substitute, then multiply: 2 × 4 + 3 × 5 = 8 + 15 = 23.**

Correct answer: 23

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Evaluate $x^2 - 4$ when $x = 5$. Student's answer: 1

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student subtracted before squaring. $x^2 = 25$, so $25 - 4 = 21$.**

Correct answer: 21

Find the Error

Grade 6 · 6.EE.3

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Expand $3(x + 4)$. Student writes: $3x + 4$

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **3 must multiply both terms inside the parentheses. The answer is 3x + 12.**

Correct answer: **3x + 12**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Simplify 5x + 2x + 7. Student writes: 10x + 7

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student added the coefficients wrong. 5x + 2x = 7x, so the answer is 7x + 7.**

Correct answer: **7x + 7**

Find the Error

Grade 6 · 6.EE.3

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Factor 12 + 8x using the greatest common factor. Student writes: 2(6 + 4x)

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **2 is a factor, but it is not the greatest. The GCF of 12 and 8x is 4, so the answer is 4(3 + 2x).**

Correct answer: **4(3 + 2x)**

Find the Error

Grade 6 · 6.EE.4

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Are $2(x + 3)$ and $2x + 3$ equivalent? Student says: Yes

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **$2(x + 3) = 2x + 6$, which is not $2x + 3$. The expressions are not equivalent.**

Correct answer: **No**

Find the Error

Grade 6 · 6.EE.4

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Are $4(x + 2)$ and $4x + 8$ equivalent? Student says: No

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **$4(x + 2) = 4x + 8$. The expressions are equivalent.**

Correct answer: **Yes**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Which expression is equivalent to 3(2x + 3)? Student says: 6x + 3

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **3 × 3 = 9, not 3. The equivalent expression is 6x + 9.**

Correct answer: **6x + 9**

Find the Error

Grade 6 · 6.EE.5

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Is x = 4 a solution of x + 7 = 11? Student says: No

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **4 + 7 = 11, which is true. So x = 4 is a solution.**

Correct answer: **Yes**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Is $x = 3$ a solution of $2x > 8$? Student says: Yes

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **$2 \times 3 = 6$, and 6 is not greater than 8. So $x = 3$ is not a solution.**

Correct answer: **No**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Is $x = 5$ a solution of $3x < 12$? Student says: Yes

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **$3 \times 5 = 15$, and 15 is not less than 12. So $x = 5$ is not a solution.**

Correct answer: **No**

Find the Error

Grade 6 · 6.EE.6

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Tickets cost $8 each. Write an expression for the cost of n tickets. Student writes: n + 8

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **Each ticket costs $8, so the cost of n tickets is found by multiplying. The expression is 8n.**

Correct answer: **8n**

Find the Error

Grade 6 · 6.EE.6

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Sam is 4 years older than Ann. Let a be Ann's age. Write an expression for Sam's age. Student writes: a - 4

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **Sam is older, so his age is more than Ann's. The expression is a + 4.**

Correct answer: **a + 4**

Find the Error

Grade 6 · 6.EE.6

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Ben has x marbles and gives away 5. Write an expression for how many he has left. Student writes: 5 - x

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **Ben starts with x and gives away 5, so the remaining amount is x − 5.**

Correct answer: **x − 5**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Solve x + 9 = 14. Student's answer: 23

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student added 9 to both sides. Subtract 9 from both sides: x = 5.**

Correct answer: 5

Find the Error

Grade 6 · 6.EE.7

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Solve 4x = 36. Student's answer: 32

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student subtracted 4 instead of dividing. Divide both sides by 4: x = 9.**

Correct answer: 9

Find the Error

Grade 6 · 6.EE.7

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Solve $x \div 5 = 7$. Student's answer: 12

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student added 5 to 7. Multiply both sides by 5: x = 35.**

Correct answer: **35**

Find the Error

Grade 6 · 6.EE.8

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Write an inequality for: x is greater than 5. Student writes: x < 5

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **Greater than means the symbol points to the smaller number. The inequality is x > 5.**

Correct answer: **x > 5**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Write an inequality for: at most 12 students. Student writes: n < 12

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **At most includes 12. The inequality is n ≤ 12.**

Correct answer: **n ≤ 12**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Solve $x - 4 > 3$. Student's answer: $x > -1$

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student subtracted 4. Add 4 to both sides: $x > 7$.**

Correct answer: **$x > 7$**

Find the Error

Grade 6 · 6.EE.9

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

The cost C is 5 times the hours t. Which variable is independent? Student's answer: C

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The cost depends on the hours. Hours is the independent variable, so the answer is t.**

Correct answer: t

Find the Error

Grade 6 · 6.EE.9

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A car travels d = 60h miles in h hours. Which is the dependent variable? Student's answer: h

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The distance depends on the hours, so d is the dependent variable.**

Correct answer: **d**

Find the Error

Grade 6 · 6.EE.9

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A plant's height is h = 2 + 3w inches after w weeks. Which is the independent variable? Student's answer: h

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The height depends on the number of weeks. Weeks is the independent variable, so the answer is w.**

Correct answer: **w**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A triangle has a base of 10 and a height of 6. What is its area? Student's answer: 60

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student forgot to multiply by $\frac{1}{2}$. Area = $\frac{1}{2}$ × 10 × 6 = 30.**

Correct answer: 30

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A parallelogram has a base of 8 and a height of 5. What is its area? Student's answer: 26

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student added the sides. Area = base × height = 8 × 5 = 40.**

Correct answer: 40

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A trapezoid has bases of 6 and 10 and a height of 4. What is its area? Student's answer: 64

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student forgot to divide by 2. Area = (6 + 10) × 4 ÷ 2 = 32.**

Correct answer: **32**

Find the Error

Grade 6 · 6.G.2

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Find the volume of a box that is $\frac{1}{2}$ unit long, 3 units wide, and 2 units tall. Student's answer: 5.5 cubic units

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student added the edges. Multiply: $\frac{1}{2}$ × 3 × 2 = 3 cubic units.**

Correct answer: **3 cubic units**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Find the volume of a box that is $2\frac{1}{2}$ units long, 2 units wide, and 3 units tall. Student's answer: 7.5 cubic units

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student left out the width. 2.5 × 2 × 3 = 15 cubic units.**

Correct answer: **15 cubic units**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Find the volume of a cube with edges of $\frac{1}{2}$ foot. Student's answer: $\frac{1}{4}$ cubic foot

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student multiplied only two edges. $(\frac{1}{2}) \times (\frac{1}{2}) \times (\frac{1}{2}) = \frac{1}{8}$ cubic foot.**

Correct answer: $\frac{1}{8}$ **cubic foot**

Find the Error

Grade 6 · 6.G.3

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A square has vertices at (0, 0), (4, 0), (4, 4), and (0, 4). What is its side length? Student's answer: 16

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: 16 is the area, not the side length. The side length is 4 units.

Correct answer: 4

Find the Error

Grade 6 · 6.G.3

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A triangle has vertices at (0, 0), (6, 0), and (0, 8). What is its area? Student's answer: 14

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student added the base and height. Area = $\frac{1}{2}$ × 6 × 8 = 24 square units.**

Correct answer: **24 square units**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A rectangle has vertices at (1, 1), (5, 1), (5, 3), and (1, 3). What is its perimeter? Student's answer: 8

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student found the area of 4 × 2 = 8. The perimeter is 4 + 2 + 4 + 2 = 12 units.**

Correct answer: **12 units**

Find the Error

Grade 6 · 6.G.4

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Find the surface area of a cube with edges of 3 cm. Student's answer: 36 square cm

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **Each face is 3 × 3 = 9 square cm. Six faces make 6 × 9 = 54 square cm.**

Correct answer: **54 square cm**

Find the Error

Grade 6 · 6.G.4

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Find the surface area of a box that is 2 by 3 by 4 units. Student's answer: 26

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student added the dimensions. The areas are 2(6) + 2(8) + 2(12) = 52 square units.**

Correct answer: **52 square units**

Find the Error

Grade 6 · 6.G.4

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Find the surface area of a box that is 5 by 2 by 1 inches. Student's answer: 17

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student forgot to count each pair of faces twice. The total is 2(10 + 2 + 5) = 34 square inches.**

Correct answer: **34 square inches**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Is 'How many pets does each student in our class have?' a statistical question? Student says: No

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **Students will give different answers, so the question has variability. It is a statistical question.**

Correct answer: Yes

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Is 'What color is the teacher's shirt today?' a statistical question? Student says: Yes

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **There is one answer for the teacher's shirt, so there is no variability. It is not a statistical question.**

Correct answer: No

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Is 'How long does it take students to walk to school?' a statistical question? Student says: No

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **Walking times vary from student to student, so it is a statistical question.**

Correct answer: **Yes**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Find the range of the data: 2, 4, 4, 6, 9. Student's answer: 9

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **9 is the largest value, not the range. The range is 9 − 2 = 7.**

Correct answer: 7

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Find the median of 3, 5, 5, 7, 8, 10. Student's answer: 5

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **With six values, the median is the mean of the two middle numbers, 5 and 7. (5 + 7) ÷ 2 = 6.**

Correct answer: 6

Find the Error

Grade 6 · 6.SP.2

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Find the mean of 10, 12, and 14. Student's answer: 10

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **10 is the smallest value, not the mean. The mean is (10 + 12 + 14) ÷ 3 = 12.**

Correct answer: 12

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Find the mean of 1, 2, 3, 4, and 15. Student's answer: 3

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **3 is the median, not the mean. The mean is (1 + 2 + 3 + 4 + 15) ÷ 5 = 25 ÷ 5 = 5.**

Correct answer: 5

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A set of 5 test scores has a mean of 7. Student's answer: The sum is 2

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student divided instead of multiplying. Sum = 7 × 5 = 35.**

Correct answer: 35

Find the Error

Grade 6 · 6.SP.3

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Four numbers have a mean of 8. What is their sum? Student's answer: 2

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student divided 8 by 4. The sum is 8 × 4 = 32.**

Correct answer: 32

Find the Error

Grade 6 · 6.SP.4

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A dot plot has 3 dots above the value 2 and 4 dots above the value 3. How many data values are there? Student's answer: 4

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student counted only the dots above 3. Each dot is one value, so there are 3 + 4 = 7 values.**

Correct answer: 7

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

In a box plot, what does the line inside the box show? Student's answer: the mean

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The line inside the box is the median, which splits the data in half.**

Correct answer: **The median**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

In a dot plot, what does each dot represent? Student's answer: the mean

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **Each dot represents one data value. It does not show the mean.**

Correct answer: **One data value**

Find the Error

Grade 6 · 6.SP.5

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A student says: For the data 8, 8, 8, the average is 8 and the spread is 0. Is the student correct?

Student says: No

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **All values are 8, so the center is 8 and there is no spread. The summary is correct.**

Correct answer: Yes

Find the Error

Grade 6 · 6.SP.5

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Test scores are 70, 75, 80, 85, and 90. The median is 80. What is the range? Student's answer: 15

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student subtracted 75 from 90. The range is 90 - 70 = 20.**

Correct answer: 20

Find the Error

Grade 6 · 6.SP.5

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A data set has values mostly between 40 and 60, with one value at 100. Which best describes the shape? Student says: symmetric

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The outlier at 100 creates a tail on the right, so the shape is skewed right, not symmetric.**

Correct answer: **Skewed right**